Sharper depth dependence for Bayesian networks stable under internal noise

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Abstract

We prove an explicit learning bound for Gaussian Bayesian tanh networks whose clipped predictions are stable under simultaneous internal disturbances. For a layered Boolean teacher of width C and depth $L-1$, a sufficient sample count is $\widetilde{O}(C^4L^7)$ when input dimension, robustness calibration, accuracy, and confidence are fixed. The guarantee is independent of learner width above C. It applies in the strong-prior limit at every fixed width and to actual finite priors at every scaling exponent $\gamma \geq 2$. The support restriction is an explicit calibrated internal-noise condition; no population-merging hypothesis is added. The proof samples neurons and rounds their weights during the virtual forward computation. Conditional independence needed for concentration survives this interleaving, while rounding becomes a controlled disturbance of the uncompressed robust network. This removes the extra depth factor from exponentially fine parameter coding. We give the complete target-cylinder, compression, posterior-tail, and statistical arguments. An entropy-based extension gives $\widetilde{O}(C^6L^{17})$ samples at every $\gamma \geq 1$, including finite mean field. It uses stated reconstruction estimates from an earlier paper. These improvements are not Lean formalized; the quartic rate at $1 \leq \gamma < 2$ remains open.

1 Introduction

A polynomial-depth learning theorem is more informative when its powers of teacher width and depth are explicit. For a prior restricted by stability under internal noise, an earlier compression argument gives a quartic teacher-width bound at sufficiently strong Gaussian scales [1]. Its joint width–depth estimate is $\widetilde{O}(C^4L^8)$. One depth factor comes from coding a compressed network to exponentially fine weight precision: a small rounding error can otherwise be amplified by every subsequent layer.

We remove that factor without changing the prior restriction. The key is to round the sampled weights *while constructing* the virtual computation. Each rounding error can then be included in the disturbances against which the original network is already stable. The sampled network need not inherit a robustness property of its own. This distinction is necessary: clean-output approximation alone does not imply that an arbitrary later parameter perturbation remains harmless.

Our result is statistical. The learner is the exact posterior, and no efficient sampling or inference algorithm is asserted. The target is a layered Boolean circuit, not an arbitrary real-weight neural teacher. The theorem applies to any common iid training and test input distribution. Appendix B transfers the same rounding principle to an entropy-based compressor and improves the all-scale depth power from eighteen to seventeen, without changing the prior support.

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2 Learner, target, and robustness restriction

Let $d \geq 1$, $L \geq 2$, and $H = L - 1$. Inputs lie in $\mathcal{X} = \{-1, 1\}^d$, with arbitrary distribution D. A learner of hidden width N has parameters $w = (W^1, \ldots, W^H, v)$ and computes

$$h^1 = \tanh(W^1 \bar{x}), \qquad h^s = \tanh(W^s h^{s-1}), \qquad z_w(x) = v^\top h^H, \qquad \bar{x} = (x, 1).$$

The raw law $\mu_{N,t}$ has independent centered Gaussian entries of variance t^2/d in W^1, t^2/N in the internal matrices, and t^2/N^2 in v. Here $0 < t \leq 1$. The scale may be any prescribed function $t = t(d, L, N)$; it need not be constant as width varies. In particular $t(N) = N^{(1-\gamma)/2}$ gives variances $N^{1-\gamma}/d$, $N^{-\gamma}$, and $N^{-\gamma-1}$.

Define weighted energy, readout energy, and clipped score by

$$\kappa_N(w) = \frac{d}{N} \left\| W^1 \right\|_F^2 + \sum_{s=2}^{H} \|W^s\|_F^2 + N \|v\|_2^2 ,$$
$$E_v(w) = N \|v\|_2^2 , \qquad T(u) = \max\{-1/2, \min\{1/2, u\}\}.$$

Disturbances are added after each activation and at the readout:

$$h^1_\eta = \tanh(W^1 \bar{x}) + \eta^1, \quad h^s_\eta = \tanh(W^s h^{s-1}_\eta) + \eta^s, \quad z_w(x; \eta) = v^\top h^H_\eta + \eta^{\text{out}}.$$

Their norm is

$$\|\eta\|_*^2 = \sum_{s=1}^{H} \frac{\|\eta^s\|_2^2}{N} + |\eta^{\text{out}}|^2.$$

Fix $c > 0$ before seeing labels. The support condition is

$$\mathcal{R}_{N,c} = \left\{ w : |T(z_w(x; \eta)) - T(z_w(x))| \leq \frac{1}{16} \text{ for all } x \in \mathcal{X}, \quad \|\eta\|_* \leq \frac{c}{\sqrt{1 + H E_v(w)}} \right\}. \tag{1}$$

Noisy hidden values are allowed outside $[-1, 1]^N$. This is a label-independent Borel condition, since the input set is finite and the response is continuous in weights and disturbances.

For a sample $S = ((X_i, f(X_i)))_{i=1}^n$ with iid $X_i \sim D$, let $Q_{S,N,t,c}$ be the raw Gaussian conditioned on

$$w \in \mathcal{R}_{N,c}, \qquad f(X_i) z_w(X_i) > 1/2 \quad (1 \leq i \leq n). \tag{2}$$

There is no initial clean-output-margin condition on unobserved inputs. The risk is posterior-average test-margin failure,

$$R(Q) = \mathbb{E}_{w \sim Q} \mathbb{P}_{X \sim D}\{f(X) z_w(X) \leq 1/4\}.$$

The target is a single-output Boolean circuit of width $C \geq 2$ and H layered stages, using NAND, COPY, and NOT. A constant-one wire is reserved at every stage and counted toward C; compilation overhead from another basis is also counted. Unused slots and stages may contain constants or copies. Thus a teacher with $h \leq H$ stages can be padded to the learner's depth. If the learner matches the teacher's depth, $L = h + 1$; if it is deeper, its actual L occurs in the bound. Width independence does not mean depth independence.

3 Explicit learning theorem

Define

$$
\begin{aligned}
e_* &= \frac{1}{16\sqrt{2C}}, \qquad \tau_0 = e_*/4, \\
b_L &= \sqrt{16\log 9 + 8\log(4L)}, \\
A &= \max\{32 b_L, 2\log(8/e_*)\}, \qquad a = \max\{2, 256c\}, \\
\rho &= \min\{1/4, A/[4(d+1)], Ae_*/(64\sqrt{C})\}, \\
q_0 &= d + 2 + (L-2)C, \\
B_0 &= 9dA^2 + 18(L-2)CA^2 + 2a^2C, \\
D_* &= B_0 + q_0 \log \frac{e\sqrt{2\pi}}{2\rho} + \log 2.
\end{aligned} \tag{3}
$$

These constants provide a target-realizing subset of the prior support of raw mass at least $\exp(-ND_*/t^2)$ at every $N \geq C$; the proof is in Appendix A. In particular $D_* \geq 1$ and every posterior in the theorem below is well-defined.

For $K \geq 1$, set

$$
\begin{aligned}
r &= \min\{1/6, c/\sqrt{1+HK}\}, & m &= \left\lceil 256 K r^{-2}[d + \log(400L)] \right\rceil, \\
B &= \sqrt{4mK}, & \zeta &= \frac{r}{4\sqrt{(d+1)^2 + Hm^2}}, \\
P &= m(d+1) + (L-2)m^2 + m, & J &= P\log(3 + 2B/\zeta).
\end{aligned} \tag{4}
$$

Theorem 1 (Uniform-width learning with polynomial coding precision)**.** *Let $0 < \varepsilon, \delta < 1$ and suppose f has the Boolean implementation specified above. The sample condition*

$$
n \geq \frac{2}{\varepsilon}\left[J + \log(1/\delta)\right] \tag{5}
$$

suffices in either of the following cases:

1. *Set $K = 8D_*$. Allow every $N \geq C$ and every positive scale satisfying*

$$
t^2 \leq \min\left\{1, \frac{ND_*}{(P_N/2)\log 2 + \log(2/\varepsilon)}\right\}, \qquad P_N = N(d+1) + (L-2)N^2 + N.
$$

2. *Set $K = 8D_* + 4(d+L) + 4\log(2/\varepsilon)$. Allow every $N \geq C$ and $0 < t \leq N^{-1/2}$, including every fixed exponent $\gamma \geq 2$.*

With probability at least $1-\delta$ over the sample, the risk $R(Q_{S,N,t,c})$ is at most ε simultaneously for all indicated widths and scales.

For every fixed width, the first clause permits all sufficiently small positive scales and therefore gives the strong-prior risk conclusion $\limsup_{t\downarrow 0} R(Q_{S,N,t,c}) \leq \varepsilon$ on the same sample event. No identification of a separately defined zero-temperature posterior is needed. If σ denotes the strong-prior parameter in the convention with density proportional to $\exp(-\kappa_N/(2\sigma^2))$, then $t = \sqrt{N}\,\sigma$.

Corollary 2 (Explicit width and depth powers)**.** *For fixed $d, c, \varepsilon, \delta$, put*

$$F = 1 + \log^2(2 + C) + \log(2 + L), \qquad G = \log(2 + CL).$$

In either scale regime of Theorem 1, a sufficient sample count has order

$$O\Big(C^4 L^7 F^4 \log^2(2 + L) G\Big) = \widetilde{O}(C^4 L^7). \tag{6}$$

At fixed depth this is $O(C^4 \log^9(2 + C))$; at fixed width it is $O(L^7 \log^7(2 + L))$.

The exact statistical dependence is in (4)–(5). In the second scale regime accuracy also enters K through $\log(2/\varepsilon)$. The asymptotic display (6) holds other resources fixed; it is not an optimality claim.

For the usual confidence criterion, define $p_S(x) = Q_{S,N,t,c}\{f(x)z_w(x) > 1/4\}$. Then $D\{p_S \le 2/3\} \le 3R(Q)$; using $\varepsilon/3$ in the theorem gives correct-margin posterior probability strictly greater than $2/3$ on all but an ε fraction of test inputs. This also gives correct-sign confidence and covers arbitrary posterior mixtures.

4 A quantized sampling construction

We prove the deterministic finite-code statement needed for the posterior. Fix N and a network w with $\kappa_N(w) \le K$. Write $k_s = \|W^s\|_F^2$ for $2 \le s \le H$ and $k_L = N\,\|v\|_2^2$. Their sum is at most K.

4.1 Pruning and conditional concentration

In every internal row and in the readout, remove entries with absolute value greater than

$$\frac{8\sqrt{K}\ \|\mathrm{row}\|_2}{r\sqrt{N}}.$$

Zero rows remain zero. Denote pruned arrays by $\bar{W}^s$ and $\bar{v}$. The absolute mass removed from a row is at most its squared norm divided by this threshold. On any vector in $[-1, 1]^N$, the internal pruning error therefore has Euclidean norm at most $r\sqrt{Nk_s/K}/8$; the readout error is at most $r\sqrt{k_L/K}/8$.

Let $S^1, \dots, S^H$ be independent lists of m uniform indices in $\{1, \dots, N\}$, with replacement. Let $\mathcal{Q}_\zeta$ round a real number to a nearest multiple of ζ, with a fixed tie convention. It preserves zero and has error at most ζ. Define the *full virtual states* by

$$\begin{aligned}
\widehat{h}_i^1(x) &= \tanh\left(\sum_{j=1}^{d+1} \mathcal{Q}_\zeta(W_{ij}^1)\bar{x}_j\right), \\
\widehat{h}_i^s(x) &= \tanh\left(\sum_{j=1}^{m} \mathcal{Q}_\zeta\left(\frac{N}{m}\bar{W}^s_{i,S_j^{s-1}}\right)\widehat{h}^{s-1}_{S_j^{s-1}}(x)\right), \quad 2 \le s \le H, \\
\widehat{z}(x) &= \sum_{j=1}^{m} \mathcal{Q}_\zeta\left(\frac{N}{m}\bar{v}_{S_j^H}\right)\widehat{h}^H_{S_j^H}(x).
\end{aligned} \tag{7}$$

The stored coordinates at the lists S^s form an ordinary width-m tanh network. This holds even when $m > N$ or indices repeat.

For each $s \geq 2$, the state $\widehat{h}^{s-1}$ depends only on $S^1, \ldots, S^{s-2}$. It remains in $[-1,1]^N$ and is independent of the fresh list S^{s-1}. Conditional on the preceding lists, for one uniform index I define

$$Y = N\bar{W}^s_{:,I}\widehat{h}^{s-1}_I.$$

The pruning threshold and bounded states give

$$\mathbb{E}\,\|Y - \mathbb{E}Y\|_2^2 \leq Nk_s, \qquad \|Y - \mathbb{E}Y\|_2 \leq 16\sqrt{NKk_s}/r.$$

Hilbert-space Bernstein [4], applied at tolerance $r\sqrt{Nk_s/K}/8$, bounds the failure probability by

$$2\exp\left(-\frac{3mr^2}{640K}\right) \leq 2\exp\left(-\frac{mr^2}{256K}\right).$$

The readout uses the independent final list S^H and has the same calculation without factors $\sqrt{N}$. Zero-energy stages are exact. A union bound over the 2^d inputs and at most L stages gives simultaneous success probability at least 199/200 by (4). The concentration estimate is applied *before* coefficient rounding at the current stage, but at the already rounded preceding virtual state. This is why its conditional independence remains valid.

4.2 All errors are disturbances of the original network

On this concentration event, pruning and sampling together give stage preactivation errors at most $r\sqrt{k_s/K}/4$ in normalized Euclidean norm, with the scalar counterpart at the readout. Tanh is 1-Lipschitz, so the corresponding stacked postactivation/readout error has norm at most $r/4$.

Rounding adds at most $(d+1)\zeta$ per first-layer coordinate, and at most $m\zeta$ per coordinate in every later layer and at the readout. This uses only that all virtual hidden states in (7) lie in $[-1,1]^N$. The stacked rounding error thus satisfies

$$\|\eta_{\text{round}}\|_*^2 \leq \zeta^2[(d+1)^2 + Hm^2] = r^2/16.$$

Define the actual disturbances in the original, unpruned network by

$$\eta^1 = \widehat{h}^1 - \tanh(W^1\bar{x}), \qquad \eta^s = \widehat{h}^s - \tanh(W^s\widehat{h}^{s-1}), \qquad \eta^{\text{out}} = \widehat{z} - v^\top\widehat{h}^H.$$

They reproduce the virtual states and output exactly. Splitting each into the pruning/sampling and rounding contributions and using the triangle inequality in the stacked norm gives

$$\|\eta\|_* \leq r/4 + r/4 = r/2. \tag{8}$$

If $w \in \mathcal{R}_{N,c}$, then $r \leq r_c(w)$ because $E_v(w) \leq K$. The original robustness assumption therefore yields

$$\sup_{x\in\mathcal{X}} |T(z_w(x)) - T(\widehat{z}(x))| \leq 1/16. \tag{9}$$

There is no subsequent unprotected rounding step, and no assertion that the sampled network itself satisfies (1).

4.3 A finite code independent of learner width

Consider the same sampled coefficients before applying $\mathcal{Q}_\zeta$. Their energy expectations obey

$$\frac{d}{m}\mathbb{E}\left\|W^1_{\mathrm{comp}}\right\|_F^2 = \frac{d}{N}\left\|W^1\right\|_F^2, \qquad \mathbb{E}\left\|W^s_{\mathrm{comp}}\right\|_F^2 \le \|W^s\|_F^2, \qquad m\mathbb{E}\,\|v_{\mathrm{comp}}\|_2^2 \le N\,\|v\|_2^2.$$

These statements concern sampled parameters, not virtual activations. They use independence of adjacent index lists and are unaffected by rounding inside (7). Markov's inequality gives sampled unrounded energy at most $4K$ with probability at least $3/4$. This event intersects the concentration event.

On the intersection, every unrounded stored coefficient has absolute value at most $B = \sqrt{4mK}$. Its rounded value belongs to a fixed grid containing at most $3+2B/\zeta$ points. Although rounding was defined on all real numbers for the full virtual computation, only these bounded stored coefficients enter its width-m realization. Thus a fixed class $\mathcal{G}$ of rounded networks satisfies

$$\log|\mathcal{G}| \le P\log(3+2B/\zeta) = J, \tag{10}$$

and every internally robust energy-K network at every width has a member of this class satisfying (9).

5 Posterior learning and the sample polynomial

Proof of Theorem 1. Let ψ equal one on $(-\infty, 1/3]$, zero on $[5/12, \infty)$, and be affine between these thresholds. Every robust network of energy at most K has a rounded representative $g \in \mathcal{G}$. If it fits a training margin, then

$$f(X_i)T(g(X_i)) \ge 1/2 - 1/16 = 7/16 > 5/12.$$

Every test input where $f(x)z_w(x) \le 1/4$ satisfies $f(x)T(g(x)) \le 1/4 + 1/16 = 5/16 < 1/3$. The representative therefore has zero empirical ramp loss, and its population ramp loss bounds the original network's test-margin error.

For a fixed nonnegative loss bounded by one with population mean u, the chance that all n iid losses equal zero is at most $(1-u)^n \le e^{-nu}$. A union bound using (10) and (5) shows that, with probability at least $1-\delta$, every zero-empirical-loss representative has population loss at most $\varepsilon/2$. This single sample event controls every robust fitting network of energy at most K, uniformly over width and scale.

The target cylinder bounds the raw conditioning probability below by e^{-ND_*/t^2} for every correctly labeled sample. Under the raw Gaussian, $N\kappa_N/t^2$ is chi-square with P_N degrees of freedom. Its exponential moment gives

$$Q_{S,N,t,c}\{\kappa_N > K\} \le \exp\left(-\frac{NK}{4t^2} + \frac{P_N}{2}\log 2 + \frac{ND_*}{t^2}\right). \tag{11}$$

For $K = 8D_*$, the first scale threshold makes this at most $\varepsilon/2$. For the second choice of K, use $t^2P_N/N \le d + L$ when $t^2 \le 1/N$. The exponent multiplied by t^2/N is at most

$$-K/4 + D_* + (d+L)(\log 2)/2 \le -\log(2/\varepsilon).$$

Since $N/t^2 \ge 1$, the tail is again at most $\varepsilon/2$. Adding this discarded posterior mass to the retained risk proves the claim for arbitrary posterior mixtures. □

Proof of Corollary 2. Put $T_L = d + \log(400L)$. The budget gives $m = O(Kr^{-2}T_L)$. All factors in B/ζ are polynomial in d, L, K, r^{-1}, so

$$J = O\Big([dKr^{-2}T_L + LK^2r^{-4}T_L^2]\log(2 + d + L + K + r^{-1})\Big).$$

The target constants imply

$$D_* = O\Big((d + LC)[\log^2(2C) + \log(2L) + \log(2 + d + C)] + (1 + c^2)C\Big).$$

At fixed $d, c, \varepsilon, \delta$, it follows that

$$K = O(CLF), \quad r^{-2} = O(CL^2F), \quad m = O(C^2L^3F^2\log(2 + L)),$$

$$P = O(C^4L^7F^4\log^2(2 + L)), \qquad \log(3 + 2B/\zeta) = O(G).$$

Substitution proves (6) and its fixed-width and fixed-depth consequences. □

6 Scope and remaining losses

The argument retains precisely the original internal-noise restriction. The improvement is in its finite code: rounding errors are absorbed during sampling rather than propagated afterward through the compressed network. The latter propagation required a log grid cost of order $L\log(2+d+L+K+r^{-1})$; here it has no leading factor of L.

The remaining depth powers come from the target cost, the calibrated noise radius, sampling width, and dense parameter counting. In particular, m is of order K/r^2 up to logarithms, and the code has order Lm^2 parameters. The posterior proof bounds readout energy by total energy, which can be of order CL up to logarithms. Removing more depth factors would require improving these estimates or using a different argument. No matching lower bound is asserted.

At mean-field scaling, expected internal Frobenius energy grows with width; for $1 < \gamma < 2$, it grows as $N^{2-\gamma}$. Thus the total-energy tail used here does not prove the quartic rate in that interval. Other compression methods give a polynomial learning theorem there [3], but that is a separate argument. No stronger support condition is introduced to bypass this limitation in the present theorem.

The target-cylinder and sampling ingredients build on the robust-prior and earlier quartic papers [2, 1], including their collaborative contributions. The proof presented here is conventional, not Lean formalized. Formal proofs for other papers in this research program do not certify this revised quantized sampling construction.

A A robust target cylinder at every width

We give the probability bound used in the statistical proof. It uses a robust digital implementation, not a claim that every implementation is stable. Let $N \geq C$. Partition the neurons into C balanced clone groups of sizes $k_a \in \{\lfloor N/C \rfloor, \lceil N/C \rceil\}$, so $C/2 \leq N/k_a \leq 2C$. Let P project onto vectors constant on each clone group. Fill unused logical slots with available constants or copies of an existing Boolean wire. The ideal wire vector $b^s(x)$ then has entries in $\{-1, 1\}$, belongs to the range of P, and has Euclidean norm $\sqrt{N}$.

Use gain A in the tanh simulation of every Boolean gate. For example, NAND on $u, v \in \{-1, 1\}$ is the sign of $1 - u - v$; copies and negations use u and $-u$. Bias constants in internal layers are supplied by counted constant wires. The ideal signed preactivation of every gate is at least A. In

the clone implementation, each coefficient from a logical wire is divided equally among that wire's clones. Set

$$e_* = \frac{1}{16\sqrt{2C}}, \qquad \tau_0 = \frac{e_*}{4}, \qquad b_L = \sqrt{16\log 9 + 8\log(4L)},$$
$$A = \max\{32b_L, 2\log(8/e_*)\}, \quad a = \max\{2, 256c\}, \quad \rho = \min\{1/4, A/[4(d+1)], Ae_*/(64\sqrt{C})\}.$$

The teacher readout averages the clones of the output wire with total coefficient a.

In Gaussian coordinates $\sqrt{d}\,W^1$, $\sqrt{N}\,W^s$, and Nv, all entries have variance t^2. Restrict each first-layer coordinate, each internal row's projection onto every normalized clone-group direction, and each readout coordinate to an interval of radius ρt about its teacher value. The internal matrices then decompose as

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^s P, \quad \|R^s\|_{\mathrm{op}} \le \rho t\sqrt{C}, \quad E^s P = 0.$$

The remaining Gaussian bulk E^s is independent of the constrained coordinates. A 1/4-net of the unit sphere has at most 9^N points, and the usual two-net Gaussian bound gives

$$\Pr\{\|E^s\|_{\mathrm{op}} > b_L t\} \le 2\exp(2N\log 9 - Nb_L^2/8) \le \frac{1}{2L}.$$

Thus all bulk norms are at most $b_L t$ with conditional probability at least 1/2.

Finite disturbance robustness. Allow a separate RMS disturbance of size τ_0 at each hidden layer and scalar output disturbance at most τ_0. This includes the entire total-budget ball $\|\eta\|_* \le \tau_0$. Write $e_s = \|h^s_\eta - b^s\|_2/\sqrt{N}$. If $e_{s-1} \le e_*$, every clone-group mean differs from its ideal bit by at most $e_*\sqrt{N/k_a} \le 1/16$. At most three teacher coefficients, including a constant coefficient, enter a gate, so its intended signed preactivation remains at least $A/2$. For $\epsilon \in \{-1, 1\}$ and real u, the elementary global secant bound

$$\epsilon q \ge A/2 \quad \Longrightarrow \quad |\tanh(q+u) - \epsilon| \le 2e^{-A/2} + 8|u|/A$$

holds: for $|u| < A/4$ use saturation, and otherwise the linear term is at least two. Since $E^s b^{s-1} = 0$ and $\|h^{s-1}_\eta\|_2/\sqrt{N} \le 1 + \tau_0$, it follows that

$$e_s \le 2e^{-A/2} + (8/A)\rho\sqrt{C}(1+\tau_0) + (8b_L/A)e_{s-1} + \tau_0 \le e_*.$$

Each of the four terms is at most $e_*/4$. The first layer initializes the same invariant: its coordinatewise preactivation perturbation is at most $\rho(d+1) \le A/4$, and its postactivation disturbance contributes at most τ_0 in RMS norm. Consequently the noisy signed readout is at least

$$f(x)z_w(x;\eta) \ge \frac{15a}{16} - \rho(1+\tau_0) - \tau_0 > 1 \qquad (x \in \mathcal{X},\ \|\eta\|_* \le \tau_0).$$

The readout center satisfies $\sqrt{N}\|v_*\| \ge a\sqrt{C/2}$, and its permitted perturbation has $\sqrt{N}\|v - v_*\| \le \rho$. Therefore $E_v \ge a^2 C/8$, whence

$$r_c(w) \le \frac{2\sqrt{2}c}{a\sqrt{C}} \le \frac{1}{64\sqrt{2C}} = \tau_0.$$

All clean and noisy clipped scores on the calibrated ball equal $f(x)/2$. The entire cylinder, with the bulk-norm event imposed, therefore belongs to $\mathcal{R}_{N,c}$ and satisfies the training likelihood for every sample. This also proves why every fixed $c > 0$ is allowed: a larger calibration requires only a polynomially larger readout coefficient.

Gaussian mass. There are Nq constrained independent scalar coordinates, where $q = d + 2 + (L-2)C$. The sum of their squared centers is at most NB_0, with

$$B_0 = 9dA^2 + 18(L-2)CA^2 + 2a^2C.$$

These loose constants cover the fan-in-two gate implementation above. For $Z \sim \mathcal{N}(0, t^2)$, $\rho \le 1/4$, and any center m,

$$\Pr\{|Z - m| \le \rho t\} \ge \frac{2\rho}{e\sqrt{2\pi}} \exp(-m^2/t^2).$$

Multiplication over the coordinates and the independent bulk event gives

$$\mu_{N,t}(\text{target cylinder}) \ge \exp(-ND_*/t^2), \qquad D_* = B_0 + q \log \frac{e\sqrt{2\pi}}{2\rho} + \log 2. \tag{12}$$

Here $D_* \ge 1$ and is bounded by a universal polynomial in d, L, C, c. The construction and estimate hold at every integer width $N \ge C$.

B Sequential quantization at every mean-field-or-stronger scale

The deterministic compressor used in the main theorem requires a bound on total network energy. At the finite mean-field scale, the unconditioned Gaussian background can have growing total energy. An entropy-based compressor avoids that difficulty. In this appendix we combine the same sequential-quantization principle with the active-direction and covariance reconstruction estimates of [3]. The result improves the unchanged internal-noise prior at every $0 < t \le 1$, including $\gamma = 1$. This appendix is not a new proof of those active-direction estimates: we state precisely which properties are imported and prove the modification that removes the coding precision's extra depth factor. Neither this modification nor its new depth exponent is Lean formalized.

B.1 Budgets and learning statement

Use the model, support $\mathcal{R}_{N,c}$, and posterior $Q_{S,N,t,c}$ from (1)–(2). Let D_* be any target-cylinder cost satisfying the conclusion of Appendix A. Thus the raw posterior conditioning event has probability at least $\exp(-ND_*/t^2)$, for every correctly labeled sample and every $N \ge C$, $0 < t \le 1$. For $0 < \varepsilon, \delta < 1$, put

$$\beta = \varepsilon/4, \qquad B_v = \max\{8, 8[D_* + \log(1/\beta)]\}, \qquad K = D_* + 1,$$
$$r = \min\{1/6, c/\sqrt{1 + (L-1)B_v}\}, \qquad \alpha = \varepsilon/12.$$

Define the compression parameters

$$e = \frac{\alpha r^2}{1000L}, \qquad \eta = \frac{\alpha}{100L}, \qquad q_a = \left\lceil \frac{2K}{e\eta} \right\rceil,$$
$$D_0 = [(d+1)q_a + 2]\log 2, \qquad V = 8K + 2q_a, \qquad B_H^2 = 4V/e,$$
$$K_0 = (4K + 2\log 2)/\eta, \qquad B_1^2 = 4[4K + 2(d+1)\log 2]/e.$$

The letter q_a denotes the number of retained active directions; it is unrelated to a population partition or a support restriction. Let m be the least integer no smaller than each of

$$10000(V + q_a + 1)/e,$$
$$64(q_a + 1)\log(4(q_a + 1)/\eta),$$
$$512K_0 r^{-2}[d\log 2 + \log(2/\eta) + 1].$$

Finally set

$$M = \max\{m, \lceil LD_0/K \rceil, 1\}, \qquad B_G = \max\{2, 2\sqrt{\log(6m/e)}\},$$

$$P_* = M(d+1) + (L-2)M^2 + M, \qquad B_{\text{small}}^2 = \frac{4(4MK + 2P_* \log 2)}{\alpha},$$

$$B_{\text{large}} = \max\left\{B_1, \frac{2B_H\sqrt{q_a+1}}{m} + \frac{\sqrt{2}B_G}{\sqrt{m}}, \frac{\sqrt{K_0}}{m}, 1\right\},$$

$$B_* = 1 + \max\{B_{\text{small}}, B_{\text{large}}, 1\}, \qquad \zeta = \frac{\sqrt{e}}{4(M+d+1)},$$

$$J_* = P_* \log(3 + 4B_*/\zeta).$$

All square roots above denote the nonnegative root.

Theorem 3 (An improved all-scale budget)**.** *Under the Boolean-teacher convention of the main theorem, the condition*

$$n \geq \frac{16}{\varepsilon}\left[J_* + \log(1/\delta)\right] \tag{13}$$

implies, with probability at least $1-\delta$ *over iid training inputs,*

$$\mathbb{E}_{w \sim Q_{S,N,t,c},\, X \sim D} \mathbf{1}\{f(X) z_w(X) \leq 1/4\} \leq \varepsilon$$

simultaneously for every $N \geq C$ *and every* $0 < t \leq 1$*. The conclusion is also valid if global clean-output-margin conditioning is added to the prior. No population-merging or additional robustness assumption is made.*

With $d, c, \varepsilon, \delta$ fixed, write

$$F = 1 + \log^2(2+C) + \log(2+L), \qquad G = \log(2+CL).$$

The target constants and the budgets above give

$$n = O(C^6 L^{17} F^6 G). \tag{14}$$

This improves the former $O(C^6 L^{18} F^6 G)$ budget while preserving its fixed-depth $O(C^6 \log^{13}(2+C))$ width consequence. A further target-cylinder sharpening below removes the $\log(2+L)$ term from F.

B.2 Imported entropy and reconstruction estimates

We use the following properties of the construction in the appendix "Finite posterior compression by internal-noise coupling" of [3]. Its lemmas on future-weight entropy, whole-layer active directions, and whitened sampling prove these properties for *any bounded virtual prefix*. They do not require a Gaussian law for the prefix and do not assume its robustness.

First, suppose a prefix is A, the current original weight block is X, and the future blocks are Y, with product Gaussian reference law $P_X \otimes P_Y$. For a finite message J, a retained quantity $H_0 = f(A, J, X)$, and a new prefix A' which is a randomized function of (A, J, H_0) using fresh randomness conditionally independent of Y,

$$\mathbb{E}_{A'} D_{\text{KL}}(Q_{Y|A'} \| P_Y) \leq \mathbb{E}_A D_{\text{KL}}(Q_{X,Y|A} \| P_X \otimes P_Y) + \mathcal{H}(J \mid A). \tag{15}$$

Here $\mathcal{H}$ denotes conditional Shannon entropy. The inequality is the KL chain rule and data processing; deterministic processing of A' cannot increase its left side.

Second, if $Q \ll \mu_{N,t}$ has $D_{\mathrm{KL}}(Q\|\mu_{N,t}) \le NK/t^2$ and $N > M$, the budgets above permit the following sequential construction. Its current virtual layer has N rows, each evaluated on m stored source neurons from the preceding layer. The first original layer is clipped rowwise at physical standardized norm B_1. Each internal stage retains at most q_a active input directions, clips the active row norms at B_H, reconstructs the complementary covariance by whitened source sampling, and clips its new Gaussian coefficients at B_G. The finite message selecting active inputs has entropy at most D_0; all other new randomness is fresh conditional on the prefix and active matrix. Hence (15) bounds the unprocessed original-weight entropy at stage s by

$$B_s \le NK/t^2 + sD_0 \le 2NK/t^2.$$

For every fixed input x, the conditional reconstruction couplings give expected squared RMS activation injection less than $20e$ per hidden stage, with success indicators. The first layer has the smaller bound e. Before any later row selection, each new internal coefficient has absolute value at most

$$\frac{2B_H\sqrt{q_a+1}}{m} + \frac{\sqrt{2}B_G}{\sqrt{m}}.$$

The readout is truncated at $N\|v\|^2 \le K_0$ and reconstructed using m source samples, with coefficient bound $\sqrt{K_0}/m$ and uniform output error at most $r/16$. The sum of probabilities of rank, Gram, and readout failures is at most $3L\eta$.

For clarity, the quantitative source-reconstruction input is the following estimate. For bounded source activations $|h_j(x)| \le 1$, active dimension at most q_a, and active matrix H_0, its whitened half-uniform, half-leverage sampling law has expected squared RMS error, with the good-Gram indicator, at most

$$\frac{2\|H_0\|_F^2/N + 16(q_a+1)}{m}.$$

The averaged active energy is bounded by V, while the Gram failure probability is at most η. These facts, Gaussian clipping, the active-direction transport estimate, and the squared triangle inequality give the stated $20e$ bound. Each bound holds for every bounded prefix to which the entropy estimate applies. These are the nontrivial reconstruction estimates imported from [3]; the argument below does not infer them merely from an existing compressed network's clean-output accuracy.

B.3 Rounding inside the canonical construction

Choose fixed scalar lattice rounding with error at most ζ and with zero fixed. After first-layer clipping, round all its coefficients before constructing any later prefix. At each internal stage apply the imported reconstruction procedure to the *already rounded* prefix, then round every newly constructed virtual row before processing the next original layer. Finally round the reconstructed readout. Keep the source rows selected at subsequent stages, as in the unrounded construction. There is no terminal rounding pass.

This modification preserves the preceding estimates. Tanh keeps the rounded virtual activations in $[-1,1]$. Each preceding state is specified before the next fresh source list is sampled. Thus the boundedness and conditional-independence assumptions of the sampling estimate remain valid. Rounding new rows is a deterministic function of the same prefix, message, active matrix and fresh Gaussian variables as their unrounded construction. It therefore preserves (15) without adding a message. The resulting prefix may have a discrete law; that law never receives a Gaussian relative-entropy comparison.

Fix an input x. The conditional transports can be glued while keeping the original full-network marginal equal to Q and the outgoing rounded-prefix/future-weight marginal equal to the next canonical stage's law. To do so, extend each transition by the previous conditional law of past original weights given its incoming prefix and remaining original weights. This disintegration preserves all earlier computed prefixes. Define the new postactivation injections by

$$\eta^s = h^s_{\text{rounded virtual}}(x) - \tanh\big(W^s_{\text{original}} h^{s-1}_{\text{rounded virtual}}(x)\big).$$

The final candidate output is exactly the original network's output under these injections and the analogous readout injection. Its marginal law is independent of x, although the coupling may depend on x.

Coefficient rounding changes a first-layer preactivation by at most $(d+1)\zeta$, and every later virtual preactivation by at most $m\zeta$. Both are at most $\sqrt{e}/4$. Consequently the expected squared RMS hidden injection, including rounding, is at most

$$2(20e) + 2(e/16) < 41e.$$

Outside the abort events, Markov's inequality gives

$$\mathbb{P}\left\{\sum_{s<L} \|\eta^s\|^2 / N > r^2/16\right\} \le \frac{656Le}{r^2} = 0.656\alpha.$$

The total abort probability is at most $3L\eta = 0.03\alpha$. The readout rounding error is at most $m\zeta \le \sqrt{e}/4 \le r/64$, where the last inequality uses $\alpha \le 1/8$ and $L \ge 2$. Together with its sampling error, the scalar injection is at most $5r/64$. On the good event,

$$\|\eta\|_* \le \sqrt{r^2/16 + 25r^2/4096} = \frac{\sqrt{281}}{64} r < r/2.$$

The exceptional probability is less than α.

When $N \le M$, the Gaussian entropy-moment inequality instead gives, with probability at least $1 - \alpha/4$, the coefficient bound B_{small} above. On this event round every original coefficient and compute the rounded network sequentially; on its complement use a zero-readout default. At each input the rounded computation is the original network under injections satisfying

$$\|\eta\|_*^2 \le \zeta^2[(d+1)^2 + (L-1)N^2] \le \zeta^2 L(M+d+1)^2 = Le/16 < r^2/4.$$

This identity uses bounded rounded hidden activations, not a bound on products of original matrix norms.

All retained coefficients, after rounding, lie in $[-B_*, B_*]$. Padding each candidate to width M with zero neurons is exact for tanh. A grid with at most $3 + 4B_*/\zeta$ entries per coefficient gives a finite class $\mathcal{H}_*$ with

$$\log|\mathcal{H}_*| \le J_*. \tag{16}$$

The construction proves the following strengthened finite-code statement: for every such Q there is one law Γ_Q on $\mathcal{H}_*$ such that, at each fixed input, a coupling has

$$h(x) = z_w(x;\eta), \qquad \|\eta\|_* < r/2$$

except on probability at most α. Only the coupling, not its two marginals, can depend on the input.

B.4 Posterior and statistical argument

Under the raw Gaussian, NE_v/t^2 is chi-square with N degrees of freedom. The target-cylinder denominator and the exponential moment bound give posterior mass at most β to $E_v > B_v$. The retained conditional posterior Q' satisfies

$$D_{\mathrm{KL}}(Q'\|\mu_{N,t}) \le ND_*/t^2 + \log(1/(1-\beta)) \le NK/t^2.$$

Every retained original network is robust on the common radius r. Apply the strengthened finite-code statement to Q'. On each good coupling the clipped-score error is at most $1/16$.

Let ψ be one on $(-\infty, 1/3]$, zero on $[5/12, \infty)$, and affine between these intervals. A correctly fitted training point has candidate ramp loss zero on the good coupling. A test-margin failure of the original network has candidate ramp loss one there. For the finite class $\mathcal{H}_*$, a relative Chernoff inequality and a union bound give, on a sample event of probability at least $1-\delta$,

$$\mathbb{E}_D\psi(f(X)T(h(X))) \le \frac{2}{n}\sum_{i=1}^{n}\psi(f(X_i)T(h(X_i))) + \frac{8[J_* + \log(1/\delta)]}{n} \qquad (h \in \mathcal{H}_*).$$

Averaging over $\Gamma_{Q'}$, comparing population risks, and restoring the discarded posterior mass bounds the desired risk by

$$\beta + 3\alpha + \frac{8[J_* + \log(1/\delta)]}{n} \le \varepsilon$$

under (13). The finite class and sample event do not depend on N, t. This proves Theorem 3. Adding global-margin conditioning changes none of these steps, since the same target cylinder is retained.

B.5 Width–depth accounting and a sharper target cylinder

For fixed $d, c, \varepsilon, \delta$, the displayed budgets give

$$\begin{aligned} D_*, K, B_v &= O(CLF), & r^{-2} &= O(CL^2F),\\ e^{-1} &= O(CL^3F), & \eta^{-1} &= O(L),\\ q_a, V &= O(C^2L^5F^2), & M &= O(C^3L^8F^3),\\ P_* &= O(C^6L^{17}F^6). \end{aligned}$$

Both B_* and ζ^{-1} are polynomial in C, L, F, so $\log(3 + 4B_*/\zeta) = O(G)$. The old terminal grid had an additional factor L here. This establishes (14).

There is a separate improvement to the target-cylinder budget which changes neither the prior nor the compression. In Appendix A, replace the depth-dependent bulk cutoff by

$$b = \sqrt{16\log 9 + 8\log 4}, \qquad A = \max\{32b, 2\log(8/e_*)\}.$$

Each unconstrained Gaussian bulk matrix has, by the same two-net bound,

$$\mathbb{P}\{\|E^s\|_{\mathrm{op}} > bt\} \le 2\exp(2N\log 9 - Nb^2/8) = 2\cdot 4^{-N} \le 1/2.$$

The bulk matrices are conditionally independent across layers. Their simultaneous success probability is therefore at least $2^{-(H-1)}$. Every deterministic stability estimate in the cylinder proof is unchanged. Recompute B_0, q_0, ρ with this constant-cutoff choice of A and use the conservative cost

$$D_*^{\mathrm{prod}} = B_0 + q_0\log\frac{e\sqrt{2\pi}}{2\rho} + H\log 2. \tag{17}$$

Indeed $N/t^2 \geq 1$ absorbs the product-event cost into this rate. At fixed d, c, with $F_0 = 1+\log^2(2+C)$, one has $D_*^{\mathrm{prod}} = O(CLF_0)$. Substituting this valid cylinder cost in Theorem 3 gives

Corollary 4 (Depth logarithms removed from the target gain)**.** *For fixed $d, c, \varepsilon, \delta$, the unchanged robust prior admits sufficient sample count*

$$O\Big(C^6 L^{17}[1+\log^2(2+C)]^6 \log(2+CL)\Big)$$

at every $N \geq C$ and every $0 < t \leq 1$. At fixed teacher width this is $O(L^{17}\log(2+L))$.

The same cylinder replacement is also available in the quartic main theorem; it is independent of the sampling and quantization argument. Its proof uses the target constants only through their valid mass lower bound. With this substitution the quartic sufficient order becomes

$$O\Big(C^4 L^7[1+\log^2(2+C)]^4 \log^2(2+L)\log(2+CL)\Big)\,.$$

At fixed teacher width this is $O(L^7\log^3(2+L))$.

Remark 5 (A conservative joint polynomial)**.** *Put $S = 2+d+L+C+c+c^{-1}+\varepsilon^{-1}+\log(1/\delta)$. The coarse accounting in [3] for the unchanged budgets gives $e^{-1} = O(S^8)$, $M = O(S^{21})$, $P_* = O(S^{43})$, and $B_* = O(S^{22})$. The new mesh has $\zeta^{-1} = O(S^{25})$. Thus $J_* = O(S^{43}\log S)$ and (13) is $O(S^{44}\log S)$, hence $O(S^{45})$. This deliberately loose degree-45 envelope is a consequence of the conventional modification proved here. It does not replace or enlarge the existing Lean certification of the earlier degree-46 envelope.*

References

[1] ChatGPT-6 Astra, Claude Fable 5 and Opus 5, Kaarel Hänni, and Dmitry Vaintrob. *A quartic-width learning bound for Bayesian networks stable under internal noise.* Research manuscript, 21 October 2026. Source and build instructions.

[2] ChatGPT-6 Astra, Claude Fable 5 and Opus 5, Kaarel Hänni, and Dmitry Vaintrob. *A polynomial-depth learning bound for NNBayes on robust networks.* Research manuscript, 18 October 2026. Source and build instructions.

[3] ChatGPT-6 Astra, Claude Fable 5 and Opus 5, Kaarel Hänni, and Dmitry Vaintrob. *A sharper polynomial learning bound for NNBayes on robust networks.* Research manuscript, 21 October 2026. Source and build instructions.

[4] Pierre Tarrès and Yuan Yao. *Online learning as stochastic approximation of regularization paths: optimality and almost-sure convergence.* Appendix A, Corollary A.2, equation (A-1). https://arxiv.org/abs/1103.5538.